

Within-session learning in DishBrain is likely explained by asymmetric feedback rather than by unpredictability

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Kagan et al.¹ showed that cortical cultures embedded in a closed-loop Pong game improve their play when a miss is followed by unpredictable stimulation and a hit by predictable stimulation, and interpreted this as the cultures acting to minimize surprise, as the free-energy principle proposes. The feedback after a miss, however, differs from the feedback after a hit not only in predictability. After a hit, all eight sensory electrodes are stimulated at 75 mV and 100 Hz for 100 ms; after a miss, random electrodes are stimulated at 150 mV and 5 Hz, a voltage chosen to be “even more disruptive to the culture,”¹ for 4 s, and play resumes only after a further 4 s. Four things change at once after a miss, the pattern, the voltage, the pulse rate and the clock, so the published design cannot say which of them teaches. The same feedback is used in the group’s recent comparison of cultures with reinforcement-learning agents,² and gameplay relative to Rest now serves as a read-out for drug effects,³ so the question bears on current work as well. The closed-loop platform and the authors’ release of per-rally data make this question answerable. We address only this question and not the terminology of intelligence and sentience, which has been debated elsewhere.^{4,5}

We re-analyzed the authors’ public gameplay data⁶ with the culture as the unit⁷ and hits per rally as the measure, comparing the first 5 minutes of gameplay (T1) with the remaining 15 (T2), as in the original study (Figure 1A). Methods are in Supplemental Information and the code is at <https://github.com/kladkogex/20-watt-computer> (folder papers/dishbrain-scrambler/reanalysis). Stimulus cultures improved within the session from 0.687 to 0.839 hits per rally (+0.152, Wilcoxon $p = 0.0002$, $n = 24$), with a positive mean gain in each of the three cell lines. Silent cultures, which receive no stimulation after a miss but the same eight-second time-out, did not improve (0.849 to 0.815, -0.033 , $p = 0.35$, $n = 17$), and neither did No feedback cultures (0.777 to 0.775, -0.002 , $p = 1$, $n = 15$). At T2 the three conditions could not be distinguished (Kruskal–Wallis $p = 0.22$, $n = 56$): Stimulus reached, within the session, the level that the other two groups had from the start. The published comparisons express gameplay relative to each culture’s Rest session, but Rest baselines differed between the groups (0.666, 0.775 and 0.787 hits per rally; Kruskal–Wallis $p = 7 \times 10^{-5}$), with the Stimulus cultures lowest. Gameplay relative to Rest reproduces the published effect (T2 minus Rest: +0.17, +0.04 and -0.01), but the difference is carried by the baselines, not by the end levels. The groups also differed in composition: Silent contains no primary mouse cultures, and No feedback contains 2 human GFP-iPSC cultures against 12 in Stimulus. In short, the cultures that received the full miss feedback climbed to where the others already were, and no group climbed higher.

There is a simpler way to get this result, and it needs no prediction at all. Think of the network’s setting ψ as a dial that is bumped after every point: each miss gives it a random, symmetric bump of variance σ_m^2 , and each hit a bump of variance σ_h^2 . For small bumps the setting diffuses with the state-dependent coefficient $D = \frac{1}{2}\sigma_m^2[\kappa + (1 - \kappa)P(\psi)]$, where P is the miss probability and $\kappa = \sigma_h^2/\sigma_m^2$, and the stationary density is $\rho \propto 1/[\kappa + (1 - \kappa)P]$ (derivation in Supplemental Information).⁸ The dial lingers wherever it is bumped least, and if misses bump harder than hits, that is wherever it

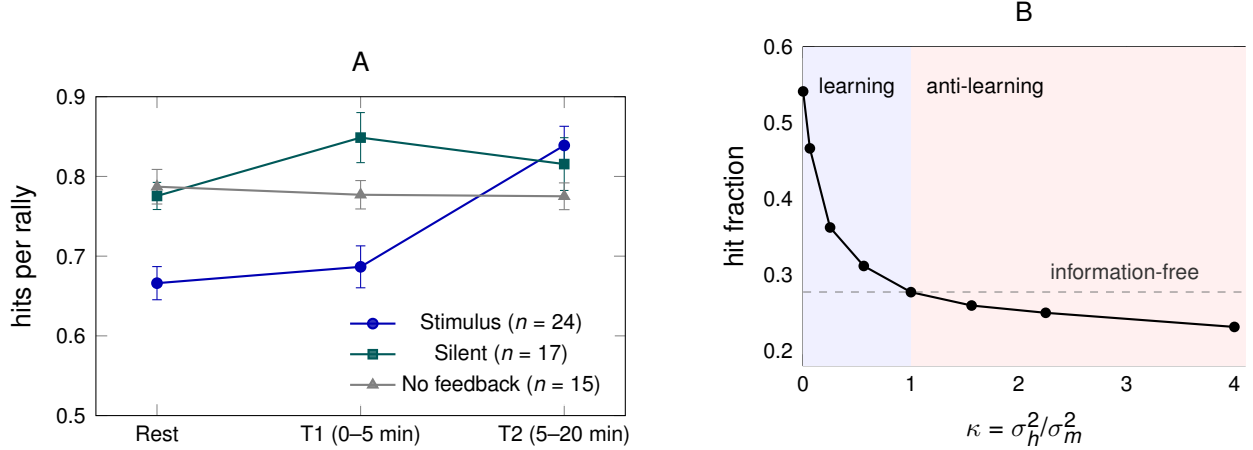


Figure 1. Re-analysis of the published data and the perturbation account. (A) Hits per rally at Rest and in the first 5 (T1) and last 15 minutes (T2) of gameplay in the three feedback conditions of Kagan et al.,^{1,6} mean $\pm$ SEM over cultures. Only Stimulus improves within the session ($p = 0.0002$); at T2 the conditions are indistinguishable ($p = 0.22$); Rest baselines differ ($p = 7 \times 10^{-5}$). (B) A closed-loop Pong model whose only teacher is a random kick to its sensorimotor weights after each approach, of variance σ_m^2 after a miss and σ_h^2 after a hit: hit fraction in the last three quarters of a session against $\kappa = \sigma_h^2 / \sigma_m^2$ (400 simulated cultures per point; the SEM over cultures is of order 0.01, comparable to the marker). Performance crosses the information-free level (dashed) at $\kappa = 1$; the model computes no prediction, reward or error.

misses least: the long-run miss rate falls below that of a random setting if and only if $\kappa < 1$, and rises above it for $\kappa > 1$ (Figure 1B). No bump is aimed; good settings are not sought, only left alone, as in the stimulus regulation principle proposed for cultures⁹ and learning by stimulation avoidance.¹⁰ Because σ_m^2 multiplies D , the size of the perturbation sets how fast a network learns but not where it ends, which is what the re-analysis shows: a fast gain under the strongest feedback and indistinguishable end levels. This account is not opposed to a free-energy description, and a surprise-minimizing system may implement exactly this stabilization, but it is the simpler mechanism the published data support, and the tests below separate the two.

The account has one gap. In a spiking model of the culture on the chip’s geometry, calibrated to the Rest recordings, two local plasticity rules without reward give $\kappa \approx 0.6$ – 0.8 , measured from each outcome to the next.⁸ The model carries no sensory-to-motor path at any size up to 10^6 neurons, because pulses at the sensory electrodes only suppress the motor regions, so it plays at chance and κ is measured under the resulting hits and misses (Supplemental Information). This contingency comes from the eight-second time-out rather than from the 150 mV pulses, because synapses drift at a nearly constant rate and a miss buys eight seconds more of it, so the model predicts learning in Silent as well, which the data do not show. Either real 150 mV stimulation perturbs synapses far more than the model’s local rules do, so that Silent, with a smaller σ_m , learns too slowly to show within a session, or Silent’s lack of improvement reflects its high starting level (0.849 at T1, above its own Rest level and above Stimulus’s end point), which neither account explains. Stimulation is also unlikely to perturb synapses equally in all directions: a 100 Hz burst can itself potentiate specific pathways (Supplemental Information).

Two protocols decide. In protocol B the miss keeps its 150 mV, 4 s and 4 s pause, but the pulses follow a fixed sequence over the eight electrodes: unpredictability predicts little or no learning, perturbation predicts learning as before. In protocol F a hit is followed by the same eight-second time-out as a miss, and only a miss is followed by 150 mV random stimulation: if the pause is what

perturbs, F does not learn; if the stimulation is, it does. Two analyses of existing recordings would also discriminate. First, if a miss moves the network's state, for example the evoked response to a fixed probe, further than a hit does, then across cultures the within-session gain should grow with that difference. Second, across Silent cultures, whose gains vary around a mean of zero, the number of spontaneous bursts after each miss should predict a culture's gain.

The two accounts fit the published conditions equally well, with the Silent qualification above, and differ on protocols B and F. We offer them as experiments that the authors' platform is well equipped to run.

Declaration of interests

The author declares no competing interests.

Supplemental information

Document S1. Methods of the re-analysis and statistics, derivation of the stationary density, and the closed-loop model.

References

1. Kagan, B.J., Kitchen, A.C., Tran, N.T., Habibollahi, F., Khajehnejad, M., Parker, B.J., Bhat, A., Rollo, B., Razi, A., and Friston, K.J. (2022). In vitro neurons learn and exhibit sentience when embodied in a simulated game-world. *Neuron* 110, 3952–3969. <https://doi.org/10.1016/j.neuron.2022.09.001>.
2. Khajehnejad, M., Habibollahi, F., Loeffler, A., Paul, A., Razi, A., and Kagan, B.J. (2025). Dynamic network plasticity and sample efficiency in biological neural cultures: a comparative study with deep reinforcement learning. *Cyborg Bionic Syst.* 6, 0336.
3. Watmuff, B., Habibollahi, F., Desouza, C., Khajehnejad, M., Loeffler, A., Baranes, K., Poulin, N., Kotter, M., and Kagan, B.J. (2025). Drug treatment alters performance in a neural microphysiological system of information processing. *Commun. Biol.* 8, 916.
4. Balci, F., Ben Hamed, S., Boraud, T., Bouret, S., et al. (2023). A response to claims of emergent intelligence and sentience in a dish. *Neuron* 111, 604–605.
5. Kagan, B.J., Razi, A., Bhat, A., Kitchen, A.C., et al. (2023). Scientific communication and the semantics of sentience. *Neuron* 111, 606–607.
6. Kagan, B.J., et al. (2022). DishBrain gameplay and Rest recordings. Open Science Framework. <https://doi.org/10.17605/osf.io/5u6qv>.
7. Aarts, E., Verhage, M., Veenvliet, J.V., Dolan, C.V., and van der Sluis, S. (2014). A solution to dependency: using multilevel analysis to accommodate nested data. *Nat. Neurosci.* 17, 491–496.
8. Kladko, K. (2026). Scrambled when wrong, left alone when right: outcome-dependent diffusion and learning in cultured neurons playing Pong. Preprint, in preparation.
9. Shahaf, G., and Marom, S. (2001). Learning in networks of cortical neurons. *J. Neurosci.* 21, 8782–8788.
10. Sinapayen, L., Masumori, A., and Ikegami, T. (2017). Learning by stimulation avoidance: a principle to control spiking neural networks dynamics. *PLoS ONE* 12, e0170388.